

EXPERIMENT-23 Family Tree

Aim

To implement a family tree in Prolog and define mother, father, grandfather, grandmother, sister, and brother relations.

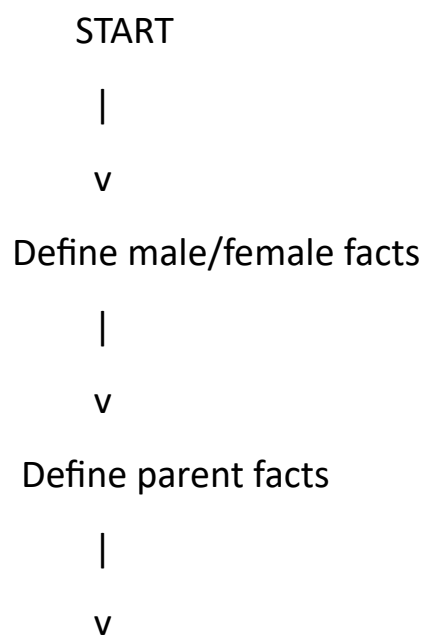
Objectives

- To represent family relationships using facts.
- To define relationships using rules.
- To perform logical queries on the family tree.

Algorithm

1. Define male and female persons.
2. Define parent relationships.
3. Define mother and father using gender and parent.
4. Define grandparents using parent relationships.
5. Define siblings based on common parents.
6. Define sister and brother based on sibling and gender.

Flowchart



Define family relations

|

v

Enter a query

|

v

Prolog searches facts

|

v

Display result

|

v

STOP

Program

% Family Tree

female(pam).

female(liz).

female(ann).

female(pat).

male(tom).

male(bob).

male(jim).

% Parent relationships

parent(pam, ann).

parent(tom, ann).

parent(pam, liz).

parent(tom, liz).

parent(ann, pat).

parent(bob, pat).

parent(ann, jim).

parent(bob, jim).

% Mother

mother(Mother, Child) :-

female(Mother),

parent(Mother, Child).

% Father

father(Father, Child) :-

male(Father),

parent(Father, Child).

% Grandfather

grandfather(Grandfather, Child) :-

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male(Grandfather),  
parent(Grandfather, Parent),  
parent(Parent, Child).
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% Grandmother

grandmother(Grandmother, Child) :-

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female(Grandmother),  
parent(Grandmother, Parent),  
parent(Parent, Child).
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% Sibling

sibling(X, Y) :-

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parent(P, X),  
parent(P, Y),  
X \= Y.
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% Sister

sister(Sister, Person) :-

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female(Sister),  
sibling(Sister, Person).
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% Brother

brother(Brother, Person) :-

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male(Brother),  
sibling(Brother, Person).
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Queries

?- mother(pam, ann).

Output:

true.

?- father(tom, liz).

Output:

true.

?- grandfather(tom, pat).

Output:

true.

?- grandmother(pam, jim).

Output:

true.

?- sister(liz, ann).

Output:

<pre>1 :- initialization(main). 2 3 % Family Tree 4 5 female(pam). 6 female(liz). 7 female(ann). 8 female(pat). 9 10 male(tom). 11 male(bob). 12 male(jim). 13 14 % Parent relationships 15 parent(pam, ann). 16 parent(tom, ann). 17 18 parent(pam, liz). 19 parent(tom, liz). 20 21 parent(ann, pat). 22 parent(bob, pat). 23 24 parent(ann, jim). 25 parent(bob, jim). 26 27 % Mother</pre>	I/O AI Agent STDIN Input for the program (Optional) Output: Mother of Ann: pam Father of Ann: tom Grandfather of Pat: tom Grandmother of Pat: pam Sister of Ann: liz Brother of Pat: jim Mother of Ann: pam Father of Ann: tom Grandfather of Pat: tom Grandmother of Pat: pam Sister of Ann: liz Brother of Pat: jim
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Result

The family tree and required relationships were successfully implemented.

Conclusion

Prolog is suitable for representing family relationships and answering relationship-based queries.